

N-Nitroso compounds in the diet.

N-Nitroso compounds were known almost 40 years ago to be present in food treated with sodium nitrite, which made fish meal hepatotoxic to animals through formation of nitrosodimethylamine (NDMA). Since that time, N-nitroso compounds have been shown in animal experiments to be the most broadly acting and the most potent group of carcinogens. The key role of nitrite and nitrogen oxides in forming N-nitroso compounds by interaction with secondary and tertiary amino compounds has led to the examination worldwide of foods for the presence of N-nitroso compounds, which have been found almost exclusively in those foods containing nitrite or which have become exposed to nitrogen oxides. Among these are cured meats, especially bacon-and especially when cooked; concentrations of 100 micrograms kg(-1) have been found or, more usually, near 10 micrograms kg(-1). This would correspond to consumption of 1 microgram of NDMA in a 100-g portion. Much higher concentrations of NDMA (but lower ones of other nitrosamines) have been found in Japanese smoked and cured fish (more than 100 micrograms kg(-1)). Beer is one source of NDMA, in which as much as 70 micrograms l(-1) has been reported in some types of German beer, although usual levels are much lower (10 or 5 micrograms l(-1)); this could mean a considerable intake for a heavy beer drinker of several liters per day. Levels of nitrosamines have been declining during the past three decades, concurrent with a lowering of the nitrite used in food and greater control of exposure of malt to nitrogen oxides in beer making. There have been declines of N-nitroso compound concentrations in many foods during the past two decades. The small amounts of nitrosamines in food are nonetheless significant because of the possibility-even likelihood-that humans are more sensitive to these carcinogens than are laboratory rodents. Although it is probable that alkyl nitrosamides (which induce brain tumors in rodents) are present in cured meats and other potentially nitrosated products in spite of much searching, there has been only limited indirect evidence of their presence. Copyright 1999 Elsevier Science B.V.